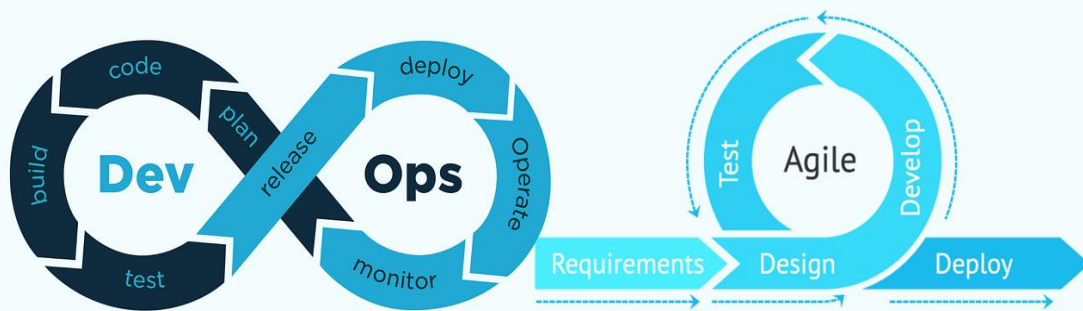


DEVOPS VS AGILE



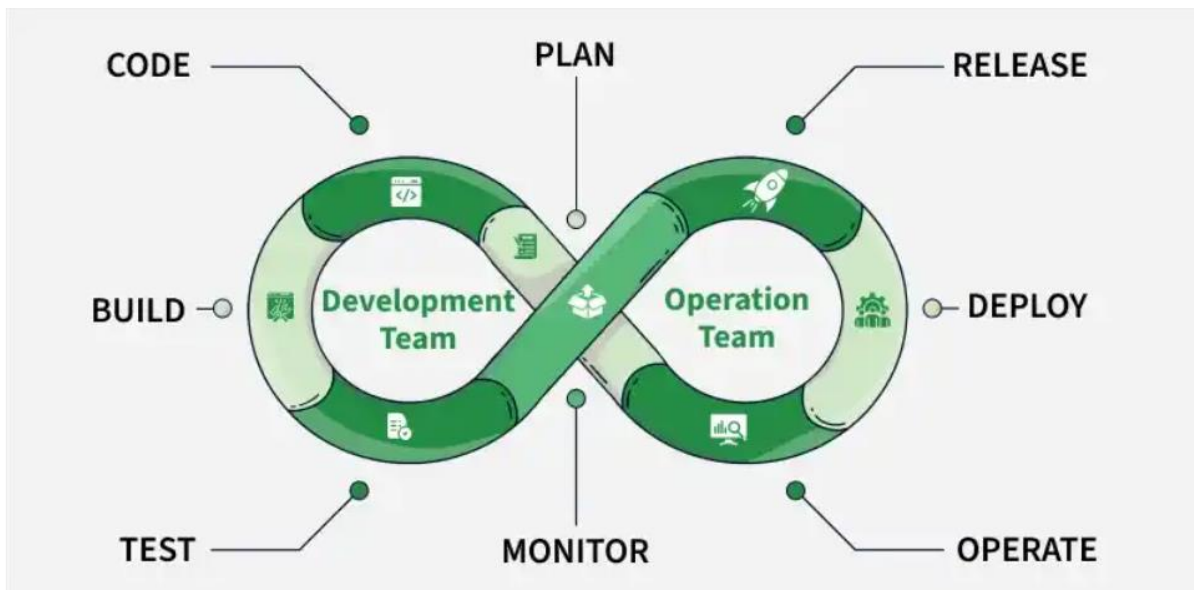
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GET – HEXAWARE TECHNOLOGIES

DATA ENGINEERING

DEVOPS



DevOps is a set of cultural philosophies, practices, and tools that unites software development (Dev) and IT operations (Ops) teams to shorten the development lifecycle and deliver high-quality software faster. By breaking down silos, it enables continuous integration/continuous delivery (CI/CD) and automation.

It focuses on collaboration, automation, and continuous improvement across the software lifecycle.

- Encourages close collaboration between development and operations teams.
- Automates build, test, and deployment processes to reduce errors.
- Enables faster and more frequent software releases.
- Improves system reliability, monitoring, and feedback loops.

Key Aspects of DevOps

- **Culture & Collaboration:** Teams share responsibility, shifting from isolated ("siloe") roles to shared ownership of the entire application lifecycle.
- **Automation:** Manual, slow processes are replaced with automated pipelines for building, testing, and releasing software.

- **Continuous Improvement:** Emphasizes fast, incremental, and continuous delivery of updates and features.
- **Core Principles (CALMS Framework):** Focuses on **C**ulture, **A**utomation, **L**ean processes, **M**easurement, and **S**haring.

Core Benefits

- **Speed:** Faster delivery of features and updates to users.
- **Reliability:** Automated testing ensures higher quality and stability.
- **Scalability:** Infrastructure is managed through code, allowing for efficient scaling.
- **Improved Collaboration:** Reduced friction between development and operations teams.

Key DevOps Practices

- **CI/CD (Continuous Integration/Continuous Deployment):** Code changes are automatically built, tested, and prepared for release.
- **Infrastructure as Code (IaC):** Managing infrastructure using configuration files rather than manual setup.
- **Monitoring and Logging:** Tracking application performance and user activity to improve reliability.
- **Incident Management:** Fast, collaborative responses to production issues.

Common Tools

- **Version Control:** Git, GitHub.
- **CI/CD:** Jenkins, GitLab CI, GitHub Actions.
- **Infrastructure & Deployment:** Terraform, Docker, Kubernetes.
- **Monitoring:** Prometheus, Datadog

Key benefits of DevOps

- **Speed and Efficiency:** Accelerated delivery through automated CI/CD pipelines, allowing teams to ship updates faster and more frequently.
- **Improved Collaboration:** Breaks down silos between development and operations teams, creating a culture of shared responsibility, trust, and improved communication.
- **Enhanced Quality and Reliability:** Automated testing and continuous monitoring ensure that code changes are vetted early, reducing bugs and leading to more stable, resilient applications.
- **Reduced Risk and Rapid Recovery:** Smaller, frequent updates combined with automated monitoring allow for faster detection, resolution, and roll-back of issues.
- **Cost Efficiency and Scalability:** Automation of manual tasks and infrastructure as code (IaC) reduces operational expenses and enables efficient scaling of resources.
- **Enhanced Security:** Integrated security practices within the development lifecycle (DevSecOps) ensure automated vulnerability scanning and compliance.
- **Increased Innovation:** By reducing time spent on maintenance and manual processes, teams have more capacity for innovation and developing new features.

Agile Methodology

Agile Methodology is a development method in which requirements and solutions evolve in cross-functional teams through collaborative effort. Its approach aligns project and product development with the customer's requirements and overall company goals. It is a methodology based on iterative development. This process encourages frequent inspection and adaptation which helps in the rapid delivery of high-quality software.

Agile methodology is developed for products and projects requiring flexibility and speed, which means almost every product and project being managed today.

Benefits of Agile

Here are the benefits of using Agile methodologies in software development:

- Flexibility and Adaptability
- Iterative Delivery
- Customer Collaboration
- Continuous Improvement
- Increased Transparency
- Enhanced Quality
- Empowered Teams
- Predictable Costs and Schedule

Scrum Methodology

Scrum is one of the most popular agile methodologies. Scrum is a lightweight, iterative, and incremental framework. Scrum breaks down the development phases into stages or cycles called "sprints". The development time for each sprint is maximized and dedicated thereby managing only one sprint at a time. The Scrum team has a scrum master and product owner with constant communication daily.

5 Scrum values

- Commitment
- Focus
- Openness
- Courage
- Respect

Benefits of scrum methodology

- Iterative Progress
- Flexibility in Requirements
- Enhanced Team Collaboration
- Clear Roles and Responsibilities
- Improved Transparency

ADVANTAGES OF AGILE

Agile methodology offers significant benefits, including faster time-to-market, enhanced product quality, and improved customer satisfaction through iterative development and continuous feedback. It increases flexibility, allowing teams to adapt to changing requirements, reduces risk, and boosts team morale through better collaboration and transparency.

Key benefits of Agile include:

- **Faster Time-to-Market:** Frequent delivery of working, high-quality software allows for faster releases.
- **High Quality:** Continuous testing, integration, and user-centric feedback loops improve product quality.
- **Customer Satisfaction:** Regular demonstration of progress and incorporating feedback ensures the product aligns with client needs.
- **Adaptability and Flexibility:** Teams can quickly pivot or change direction in response to shifting requirements.
- **Reduced Risk:** Short, iterative sprints minimize the risk of total project failure.
- **Transparency and Collaboration:** Daily stand-ups and regular reviews foster better communication and shared understanding among stakeholders.
- **Increased Productivity:** Structured, prioritized backlogs improve efficiency.
- **Increased Team Morale:** Team autonomy and better work-life balance lead to improved motivation

Product Owner and Scrum Master

In Agile Scrum, the Product Owner (PO) and Scrum Master are distinct, complementary leadership roles focused on maximizing product value and team effectiveness, respectively. The **PO defines "what"** to build by managing the backlog, while the **Scrum Master ensures "how"** it is built by facilitating processes and removing obstacles.



Product Owner (PO): The Value Maximizer

The Product Owner represents the customer and stakeholders, serving as the voice of the user to the development team and is responsible for the output or the final product ensures that it is up to date in terms of priorities with no backlog and has the items which are demanded in the current vision. The product owner represents the business and is accountable for collaborating with the customers to define what features will be in the product release.

Key Roles Include:

- **Backlog Management:** Defines, prioritizes, and maintains the product backlog (user stories, features).
- **Vision & Goal Setting:** Defines product goals and sets the release strategy.
- **Decision Authority:** Makes the final call on which features are built and in what order to maximize ROI.
- **Stakeholder Alignment:** Collaborates with customers to gather feedback and align on priorities.

Scrum Master: The Facilitator & Coach

The Scrum Master is a servant-leader focused on the health of the team and the application of Scrum practices. Scrum master is the mentor coach or the leader for the agile development teams and allows the team to self-organize. It manages the process of information exchange between the team members and takes necessary steps accordingly owing to agile principles.

Key Roles include:

- **Impediment Removal:** Clears roadblocks (technical issues, personnel blockers) to keep the team productive.
- **Process Coaching:** Educates the team and organization on Agile/Scrum principles to improve efficiency.
- **Event Facilitation:** Organizes and facilitates Sprint events (Planning, Daily Scrum, Review, Retrospective).
- **Fosters Self-Organization:** Helps the team become self-managing and high-performing.